POWER PULSE

Unsymmetrical Faults Important Formulae

Notes by BALAJI SIR

1 Single line-to-ground (LG)

1. $$I_{a0} = I_{a1} = I_{a2} = \frac{E_a}{Z_1 + Z_2 + Z_0 + 3Z_f}, \qquad I_f = 3I_{a0}$$

sequence networks in series

2 Line-to-line (LL)

2. $$I_{a1} = -I_{a2} = \frac{E_a}{Z_1 + Z_2 + Z_f}, \quad I_{a0} = 0, \quad I_b = -I_c = -j\sqrt{3}\, I_{a1}$$

networks in parallel; no zero-sequence current

3 Double line-to-ground (LLG)

3. $$I_{a1} = \frac{E_a}{Z_1 + \dfrac{Z_2(Z_0 + 3Z_f)}{Z_2 + Z_0 + 3Z_f}}, \quad I_{a0} = -I_{a1}\frac{Z_2}{Z_2 + Z_0 + 3Z_f}$$

4. $$I_f = 3I_{a0} \quad \text{(ground current)}$$

4 Three-phase and comparison

5. $$I_{3\phi} = \frac{E_a}{Z_1 + Z_f}$$

6. $$\frac{I_{LG}}{I_{3\phi}} = \frac{3Z_1}{Z_1 + Z_2 + Z_0} \quad (Z_f = 0)$$

LG exceeds 3-φ when $Z_0 < Z_1$ (with $Z_1 = Z_2$)

7. $$V_{a1} = E_a - I_{a1}Z_1, \quad V_{a2} = -I_{a2}Z_2, \quad V_{a0} = -I_{a0}Z_0$$

8. Fault current: $I_{3\phi}$ usually max for line faults; I_{LG} near generators/solidly grounded

9. With a solidly grounded neutral: $Z_f = 0,\ Z_n = 0$